Cross-Lingual Correctness Probe Transfer Reveals Architecture-Dependent Depth Effects: Distributional Matching Fails but Linear Projection Succeeds

Anonymous
anonymous@example.com

Abstract

Probe-based calibration methods extract internal correctness directions from LLM hidden states, achieving strong results in English. Whether these directions transfer cross-lingually remains untested at scale. We present the first large-scale study of correctness probe transfer across three architectures (Qwen3-235B MoE, Llama 4 Scout MoE, Gemma 4 31B Dense) and 20 languages spanning four resource tiers. We find: (1) a depth–transferability Pareto tradeoff that is architecture-dependent—9/20 languages show significant negative depth–transfer correlation on Qwen3 versus 5/20 on Llama 4; (2) CKA analysis reveals a Simpson's Paradox where raw CKA–transfer correlation is negative but partial correlation controlling for depth is significantly positive in 2/3 architectures; (3) distributional matching via CACLR (MMD-based consistency loss) is inert across a full β sweep, with CORAL and high-dimensional MMD variants also failing—geometric analysis shows these methods reduce marginal distance but leave the correctness subspace angle unchanged; (4) linear projection with $\sim$80 parallel pairs achieves 95% of maximum improvement and is the most effective label-free transfer method, particularly for low-resource languages (+5–6 pp). We release probing controls showing 22% of language–layer combinations contain capacity artifacts, concentrated in low-resource shallow layers—a methodological warning for the field.

1 Introduction

Non-English languages suffer substantially worse calibration in multilingual LLMs, with expected calibration error (ECE) reaching 5$\times$ that of English Zhou et al. (2025). Recent work demonstrates that internal correctness directions—linear features in hidden-state space that predict answer correctness—enable effective probe-based calibration for English Miao et al. (2026); Joshi et al. (2025); Anonymous (2026). These correctness directions are orthogonal to verbalized confidence Miao and Ungar (2026), occupy a low-dimensional subspace Anonymous (2026), and can be extracted with simple logistic regression Joshi et al. (2025). Yet no prior work systematically tests whether this research line extends beyond English.

The closest prior effort, Xiao et al. (2025), transfers *knowledge-boundary* probes (input-level answerability) across 8 languages, finding partial success with mean-shift alignment. However, knowledge-boundary and correctness probes target fundamentally different signals: answerability is input-dependent and binary, while correctness depends on the model's actual output and varies with question difficulty. Civelli et al. (2026) show that cross-lingual and monolingual optimal layers diverge, but do not study correctness probing or calibration. The intersection of cross-lingual transfer, correctness directions, and calibration remains open.

We present the first large-scale study of cross-lingual correctness probe transfer, spanning three architectures (Qwen3-235B, Llama 4 Scout, Gemma 4 31B), 20 languages across four resource tiers, and multiple transfer methods. Our core finding is that **deeper layers generally improve both English discrimination and cross-lingual transfer, but the deepest layers exhibit a local Pareto tradeoff where marginal English gains trade against transfer quality**—and the shape of this tradeoff is architecture-dependent. A Simpson's Paradox in the CKA–transfer relationship provides geometric

insight: raw correlation is negative (depth confound), but within each layer, higher cross-lingual CKA predicts better transfer.

Our contributions are: (1) discovery of the architecture-dependent Pareto frontier across 3 architectures and 20 languages; (2) a systematic negative result showing that distributional matching (MMD, CORAL, high-dimensional variants) fails for correctness probe transfer, with geometric explanation via CKA subspace analysis; (3) characterization of linear projection as the practical label-free solution, with resource-tier-stratified saturation at $\sim$80 parallel pairs; (4) first application of probing controls Hewitt and Liang (2019) to cross-lingual transfer, revealing that 22% of language–layer combinations are capacity artifacts.

2 Related Work

Probe-based calibration. Joshi et al. (2025) extract calibration directions from residual streams, finding strong within-model transfer across datasets but failure across models. Miao et al. (2026) discover distributed correctness representations via SAE analysis and propose cross-task steering. Miao and Ungar (2026) show correctness and verbalized confidence directions are orthogonal ($\cos < 0.04$), motivating probe-based over output-based calibration. Anonymous (2026) demonstrate that correctness occupies a 3–8D subspace accessible to linear probes. All are English-only.

Cross-lingual representations. Cross-lingual representations in multilingual LLMs exhibit linear structure amenable to projection-based alignment Conneau et al. (2018); Artetxe et al. (2018). Pires et al. (2019) and Wu and Dredze (2019) establish that mBERT transfers cross-lingually via shared subspaces. Ravisankar et al. (2026) study cross-lingual representation alignment more broadly. Our linear projection is conceptually analogous to VecMap/MUSE applied at sentence-level correctness representations; the contribution is demonstrating effectiveness for correctness probing and characterizing sample efficiency.

Multilingual calibration. Zhou et al. (2025) propose per-language layer ensembles (LACE), reducing non-English ECE from 22% to 3%, but require per-language labeled data. Xiao et al. (2025) transfer knowledge-boundary probes across 8 languages using mean-shift. We target a different signal (correctness), a different metric (calibration), more languages (20+), and analyze the Pareto frontier.

Domain adaptation theory. Ben-David et al. (2010) provide the H-divergence framework predicting that representation alignment aids transfer. Zhao et al. (2019) show that marginal distribution matching fails when conditional distributions differ—directly grounding our CACLR negative result.

Probing methodology. We follow Hewitt and Liang (2019)'s control-task framework to validate that probe AUROC reflects genuine correctness directions rather than capacity artifacts.

3 Method

3.1 Correctness Probes

Given a multilingual LLM and a multiple-choice question q with model answer a, we extract the hidden-state representation $\mathbf{h}_l \in \mathbb{R}^d$ at layer l corresponding to the last token position. A linear probe $f_l(\mathbf{h}) = \sigma(\mathbf{w}_l^\top \mathbf{h} + b_l)$ is trained on English data to predict binary correctness ($y = \mathbb{1}[a = a^*]$). We use logistic regression (LR) trained with binary cross-entropy, validated by comparison with 2-layer MLP probes (Section 4.8).

3.2 Transfer Methods

We evaluate five transfer strategies for applying English-trained probes to target language t:

Zero-shot (ZS). Apply the English probe directly: $\hat{p}_t = f_l(\mathbf{h}_l^t)$.

Mean-shift. Shift target representations by the mean difference: $\hat{p}_t = f_l(\mathbf{h}_l^t + \boldsymbol{\mu}^{\text{en}} - \boldsymbol{\mu}^t)$. Under StandardScaler + logistic regression, this adds a constant offset to all logits; since AUROC is rank-invariant, $\Delta\text{AUROC} \equiv 0$ mathematically.

CACLR. We train the probe with a composite loss:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \alpha \cdot \mathcal{L}_{\text{SB-ECE}} + \beta \cdot \mathcal{L}_{\text{MMD}} \tag{1}$$

where $\mathcal{L}_{\text{MMD}}$ uses a Gaussian RBF kernel with median-heuristic bandwidth, computed on 1-D probe logits from English and target-language parallel inputs (GlobalMMLU items). This tests whether distributional matching can improve transfer. Implementation: Adam optimizer (lr=5×10^{-4}, cosine decay to 10^{-5}), 50 epochs, early stopping (patience=10 on validation BCE), warm-started from the standard LR solution, batch size 256, 3 seeds.

Linear projection. Learn a Procrustes rotation $\mathbf{W}^* = \arg\min_{\mathbf{W}^\top\mathbf{W}=\mathbf{I}} \|\mathbf{W}\mathbf{H}^t - \mathbf{H}^{\text{en}}\|_F$ from n parallel sentence pairs, then apply: $\hat{p}_t = f_l(\mathbf{W}^*\mathbf{h}_l^t)$.

Few-shot. Retrain the probe on k target-language labeled examples ($k \in \{25, 100\}$), requiring target-language correctness labels unlike the above methods.

3.3 CKA Analysis

We compute linear Centered Kernel Alignment Kornblith et al. (2019) between English and target-language representations at each probed layer: $\text{CKA}(\mathbf{H}_l^{\text{en}}, \mathbf{H}_l^t)$. To disentangle the depth confound, we compute partial Spearman correlations between CKA and transfer AUROC controlling for layer depth.

3.4 Probing Controls

Following Hewitt and Liang (2019), we train control probes that predict random binary labels, measuring selectivity as $\text{sel} = \text{AUROC}_{\text{real}} - \text{AUROC}_{\text{control}}$. Language–layer combinations with $\text{sel} < 0.05$ are flagged as potential capacity artifacts.

3.5 Data and Models

Benchmarks. GlobalMMLU (primary, 20 languages), Belebele (reading comprehension), MMLU-ProX (translated MMLU). GlobalMMLU test split is partitioned into probe_train (70%) and eval (30%) via stratified split.

Models. Qwen3-235B-A22B (94 layers, MoE, INT4), Llama 4 Scout (48 layers, MoE, W4A16), Gemma 4 31B (60 layers, Dense, BF16). Quantization validated in Section 4.11.

Languages. 20 languages across 4 tiers: HIGH (zh, es, fr, de, pt, ja), MID (ar, ru, ko, hi, vi, tr, id), LOW (sw, bn, te, pl, nl, uk, yo).

4 Experiments

4.1 Pareto Frontier Across Architectures

We probe every 4th layer across all three architectures, computing English AUROC and zero-shot transfer AUROC for 20 languages.

Metric	Qwen3 (MoE)	Llama4 (MoE)	Gemma4 (Dense)
Layers	94	48	60
Active params	22B	17B	31B
EN best AUROC	0.81	0.77	0.79
Mean ZS AUROC	0.668	0.648	0.660
Pareto sig. (/20)	9	5	7
Median Pareto ρ	-0.38	-0.25	-0.32
CKA cliff slope	-0.022	-0.013	-0.014
EN–transfer gap	2.5 pp	1.5 pp	1.8 pp

Table 1: Architecture comparison on Pareto frontier strength. Pareto sig. = number of languages with significant negative depth–transfer Spearman ρ ($p < 0.05$).

Tier	Lang	ZS	Best	ρ	Pattern
HIGH	zh	.738	L68	-0.48*	Tradeoff
HIGH	es	.745	L72	-0.44*	Tradeoff
HIGH	fr	.740	L72	-0.42*	Tradeoff
HIGH	de	.725	L64	-0.51*	Tradeoff
HIGH	ja	.710	L64	-0.55*	Tradeoff
MID	ru	.705	L68	-0.46*	Tradeoff
MID	id	.690	L68	-0.43*	Tradeoff
MID	ar	.695	L72	-0.22	Weak
MID	ko	.688	L68	$+0.08$	None
LOW	pl	.698	L68	-0.45*	Tradeoff
LOW	nl	.720	L68	-0.47*	Tradeoff
LOW	yo	.535	L44	$+0.12$	None

Table 2: Selected per-language results for Qwen3-235B (EN-optimal L80). ZS = zero-shot AUROC at EN-optimal layer. * denotes $p < 0.05$.

Qwen3-235B. English-optimal layer is L80 (85% depth, AUROC=0.81). Zero-shot transfer at this layer averages 0.668 across 20 languages, with a clear resource-tier gradient (HIGH: 0.728, MID: 0.678, LOW: 0.582). Per-language Spearman ρ between depth and transfer AUROC reveals architecture-dependent Pareto structure: 9/20 languages show significant negative ρ ($p < 0.05$), 5 trending ($p < 0.10$), and 6 non-significant or positive. The transfer-optimal layer is L68 (72% depth), 2.5 pp below the English-optimal layer.

Llama 4 Scout. English-optimal at L44 (92% depth, AUROC=0.77). The Pareto tradeoff is weaker: only 5/20 languages show significant negative ρ (median $\rho = -0.25$ versus Qwen3's -0.38). The transfer-optimal layer is L36 (75% depth), with a smaller EN–transfer gap of 1.5 pp.

Gemma 4 31B (Dense). English-optimal at L52 (87% depth, AUROC=0.79). As the dense architecture control, Gemma 4 shows intermediate Pareto strength: 7/20 languages with significant negative ρ (median -0.32). This confirms the Pareto tradeoff is not a MoE artifact.

Table 1 summarizes the three-architecture comparison.

Per-language patterns. Table 2 shows detailed per-language statistics for Qwen3-235B. Languages with high training-data representation (zh, es, fr) show clear Pareto tradeoffs, while ar and ko show no tradeoff or positive depth effects. The pattern is consistent with MoE routing creating language-specific expert subsets at depth.

Arch.	Raw ρ	Partial ρ	p	Sig.?
Qwen3-235B	-0.24	$+0.16$	.072	No
Llama4 Scout	-0.30	$+0.31$	.012	Yes
Gemma4 31B	-0.26	$+0.24$	.038	Yes
Pilot Llama8B	-0.16	$+0.41$	.006	Yes
Pilot Qwen7B	-0.16	$+0.18$	.274	No

Table 3: Simpson's Paradox: raw vs. partial Spearman ρ between CKA and transfer AUROC, controlling for layer depth. Partial correlation is positive in all cases, significant in 2/3 full-scale architectures.

Tier	Qwen3	Llama4	Gemma4	ZS AUROC
HIGH	$11°$	$13°$	$11°$	0.72
MID	$22°$	$26°$	$24°$	0.66
LOW	$46°$	$41°$	$43°$	0.57

Table 4: Mean subspace angle between English and target-language correctness directions by resource tier. Larger angles correspond to worse zero-shot transfer.

Divergent languages across architectures. Several languages exhibit divergent Pareto patterns between Qwen3 and Llama 4: ar shows weak negative ρ on Qwen3 (-0.22, $p = 0.30$) but positive on Llama 4 ($+0.18$, $p = 0.58$); vi shows trending negative on Qwen3 (-0.38, $p = 0.07$) but positive on Llama 4 ($+0.15$, $p = 0.64$). These divergences likely reflect differences in training data composition and tokenizer coverage, though with $n = 2$ architectures, causal attribution is not possible.

4.2 CKA Geometry and Simpson's Paradox

Linear CKA between English and target-language representations declines with depth across all three architectures, but with different profiles: Qwen3 shows a plateau-then-cliff pattern (CKA stable at $\sim$0.80 through 68% depth, then dropping to 0.09 at the final layer), while Llama 4 and Gemma 4 show more gradual linear decline.

Simpson's Paradox. The raw Spearman correlation between CKA and transfer AUROC is significantly *negative* across all architectures ($\rho = -0.24$ to -0.30, all $p < 0.001$): languages with higher CKA appear to transfer worse. However, this is a depth confound—deeper layers simultaneously decrease CKA and increase English AUROC, creating a spurious negative correlation. Controlling for depth via partial correlation reverses the sign (Table 3).

In architectures with near-linear CKA depth gradients (Llama 4, Gemma 4), the partial correlation is significant ($\rho = 0.24{-}0.31$, $p < 0.05$), confirming that CKA provides genuine explanatory power beyond depth. In Qwen3-235B, the step-function-like CKA profile limits linear partial correlation effectiveness ($\rho = 0.16$, $p = 0.072$); within-layer analysis confirms the positive direction (17/24 layers with positive ρ).

Subspace angle analysis. We measure the angle between English and target-language correctness directions (Table 4). HIGH-resource languages show near-alignment ($10{-}13°$), while LOW-resource languages show substantial misalignment ($41{-}46°$). This provides the geometric basis for why zero-shot transfer works for high-resource but fails for low-resource languages.

4.3 CaCLR: A Systematic Negative Result

CACLR tests whether distributional matching can improve cross-lingual transfer. We sweep $\beta \in \{0.001, 0.01, 0.1, 1.0\}$ across 20 languages on Qwen3-235B (L68), with 3 seeds each.

β	**Mean AUROC**	**Δ (pp)**	**95% CI**	d
0 (ZS)	0.668	—	—	—
0.001	0.669	+0.1	[−0.5, +0.7]	0.02
0.01	0.671	+0.3	[−0.4, +1.0]	0.06
0.1	0.667	−0.1	[−0.8, +0.6]	0.03
1.0	0.661	−0.7	[−1.4, +0.0]	0.15

Table 5: β sweep for CACLR (Qwen3-235B, L68, mean over 20 languages, 3 seeds). All effects are negligible.

Method	**AUROC**	**Δ (pp)**	p
Zero-shot	0.668	—	—
MMD (1-D logit)	0.671	+0.3	.38
MMD (PCA d=50)	0.673	+0.5	.28
MMD (PCA d=100)	0.670	+0.2	.51
CORAL (1-D logit)	0.672	+0.4	.32
CORAL (PCA d=50)	0.671	+0.3	.40
Linear projection	**0.698**	**+3.0**	<.001
Few-shot 100	**0.718**	**+5.0**	<.001

Table 6: Distributional matching ablation. All marginal-matching methods are inert; linear projection and few-shot, which directly address subspace rotation, succeed.

β sweep. Table 5 shows the consistency loss is inert: the best configuration ($\beta = 0.01$) achieves +0.3 pp over zero-shot ($p = 0.38$, Cohen's $d = 0.06$). At $\beta = 1.0$, the loss begins to degrade performance (-0.7 pp). All bootstrap 95% CIs cross zero.

Distributional matching ablation. Table 6 extends the failure beyond MMD. CORAL (second-order alignment) and high-dimensional MMD (PCA $d = 50, 100$) on activation space are equally inert. The failure is not MMD-specific—it reflects a fundamental limitation of distributional matching for this problem class.

Geometric explanation. CKA subspace analysis (Table 7) reveals the mechanism: CACLR successfully reduces MMD distance (marginal alignment) but leaves the correctness subspace angle essentially unchanged. MMD matches marginal distributions but cannot detect or correct the directional rotation of the correctness subspace between languages. Linear projection succeeds precisely because it directly learns this rotation.

Component ablation. Ablating $\mathcal{L}_{\text{SB-ECE}}$ (α sweep): increasing α reduces ECE (-1.8 pp at $\alpha = 1.0$) but degrades AUROC (-0.5 pp). BCE alone (AUROC=0.668, ECE=0.19) versus full CACLR (AUROC=0.669, ECE=0.174)—the calibration component helps marginally, but the consistency component does not.

4.4 Transfer Method Comparison

Table 8 presents the full transfer method comparison on Qwen3-235B at the transfer-optimal layer (L68).

Mean-shift is mathematically inert ($\Delta\text{AUROC} \equiv 0$ under StandardScaler + LR, since adding a constant to all logits preserves ranks). CACLR is statistically indistinguishable from zero-shot. Linear projection with 200 pairs achieves +3.0 pp ($p < 0.001$), and few-shot with 100 labels achieves +5.0 pp ($p < 0.001$).

Tier	Angle before	MMD before	MMD after	Angle after
HIGH	11°	0.018	0.008	10.5°
MID	23°	0.042	0.018	22°
LOW	44°	0.105	0.048	42°

Table 7: Why distributional matching fails: CACLR reduces MMD distance but correctness subspace angle is unchanged. The conditional structure differs across languages; marginal matching cannot fix conditional mismatch Zhao et al. (2019).

Method	AUROC	Δ	p	Req.
Zero-shot	.668	—	—	0
Mean-shift	.668	0.0	n/a	0
CACLR (β=0.01)	.671	+0.3	.38	∼10K par.
Linear proj. (50)	.684	+1.6	.012	50 pairs
Linear proj. (200)	.698	+3.0	<.001	200 pairs
Few-shot 25	.685	+1.7	.008	25 labels
Few-shot 100	.718	+5.0	<.001	100 labels

Table 8: Transfer method comparison (Qwen3-235B, L68, mean over 20 languages). Linear projection is the best label-free method; few-shot with 100 labels achieves the highest absolute performance.

4.5 Linear Projection: Sample Efficiency and Effect Reversal

Saturation curves. We vary the number of parallel pairs from 10 to 200 (Table 9). Overall, 80 pairs achieves 95% of maximum improvement. Resource-tier stratification is clear: HIGH languages saturate at $\sim$38 pairs (subspace angle $\sim 11°$), while LOW languages require $\sim$95 pairs (angle $\sim 43°$).

Procrustes effect reversal. Linear projection does not uniformly help. Table 10 shows that languages with zero-shot AUROC > 0.65 are *harmed* by projection (-1.5 pp mean), while those with AUROC < 0.50 benefit substantially (+5.5 pp). The reversal threshold is $\sim$0.57 (95% CI: $[0.48, 0.64]$). When the correctness direction is already near-aligned ($< 15°$), the estimation noise in the Procrustes rotation exceeds the alignment gain.

4.6 Difficulty Decomposition

We test whether cross-lingual transfer decomposes into difficulty-shared and capability-specific components. IRT difficulty probes ($R^2 = 0.32$) extract a difficulty direction; projecting it out and retraining correctness probes yields the results in Table 11.

Projecting out the difficulty direction generally improves transfer for mid-resource languages (+2–3 pp) but harms low-resource languages where difficulty and capability are more entangled. The mean effect (+1.8 pp) exceeds the random-direction control (+0.7 pp) by a factor of 2.6, confirming that $\sim 60\%$ of the effect is difficulty-specific. However, prescriptive accuracy (predicting which strategy benefits a given language) is only 56%, barely above chance. We therefore present this as a **descriptive** finding: the decomposition explains what transfer contains but does not reliably predict optimal strategy.

4.7 Cross-Lingual Orthogonality

The correctness–confidence orthogonality discovered by Miao and Ungar (2026) for English ($\cos < 0.04$) persists across languages. Mean cosine similarity is 0.04 overall, with 17/20 languages below 0.10 and 13/20 below 0.05. HIGH-resource languages closely replicate the English result (mean $\cos = 0.026$), while LOW-resource languages show higher cosine (mean 0.075), driven by three outliers (am: 0.148, yo:

Pairs	HIGH	MID	LOW	Overall
10	+0.1	+0.8	+2.5	+1.1
20	+0.3	+1.5	+3.8	+1.7
50	+0.4	+2.3	+5.2	+2.5
100	+0.5	+2.6	+5.8	+2.9
200	+0.5	+2.7	+6.2	+3.0

Table 9: Linear projection ΔAUROC (pp) by resource tier and parallel pair count (Qwen3-235B, L68). LOW-resource languages benefit most.

ZS AUROC band	n	Mean Δ	Example
< 0.50	3	+5.5 pp	am, yo, te
$0.50-0.58$	3	+3.2 pp	sw, bn, tr
$0.58-0.62$	2	+1.2 pp	hi, vi
$0.62-0.67$	4	+0.2 pp	ar, ko, id, uk
$0.67-0.72$	4	-0.8 pp	ru, ja, de, pt
> 0.72	4	-1.8 pp	zh, es, fr, nl

Table 10: Procrustes effect reversal: linear projection helps low-ZS languages but harms high-ZS ones. Deployment recommendation: use projection when ZS AUROC < 0.57, skip for > 0.65.

0.122, te: 0.105). Probing controls (Section 4.10) confirm these outliers have low selectivity, indicating the elevated cosine is a noise artifact from weak directional signals rather than genuine alignment. The two architectures show no significant difference (0.037 vs 0.043, $p = 0.42$), indicating orthogonality is a structural property of the representation space.

4.8 Probe Complexity Validation

We compare logistic regression (LR) and 2-layer MLP probes to validate our methodological choice (Table 12).

MLP probes achieve +2.1 pp higher English AUROC by exploiting English-specific activation patterns, but these do not transfer: the cross-lingual gap widens from $\sim$13 pp (LR) to $\sim$18 pp (MLP). LR's single-hyperplane constraint forces it to learn language-agnostic features, making it the correct choice for cross-lingual transfer.

4.9 Cross-Benchmark Generalization

Table 13 validates our findings on Belebele (reading comprehension) and MMLU-ProX (translated MMLU).

Belebele shows weaker Pareto (6 vs 9 significant languages) as expected for passage-dependent correctness, but the Pareto structure is preserved. CACLR remains inert on both benchmarks. Linear projection remains effective. Cross-benchmark language ranking correlation is $\rho = 0.89$, confirming results are not benchmark-specific.

4.10 Probing Controls

Following Hewitt and Liang (2019), we train control probes on all 20 languages $\times$ 3 architectures $\times$ all probed layers. Table 14 shows the artifact rate by resource tier and depth.

Overall, 22% of language–layer combinations have selectivity < 0.05, concentrated in LOW-resource shallow layers (80% artifact rate at 0–25% depth). At the English-optimal depth, only am, yo, and te fall below the selectivity threshold (selectivity 0.015–0.054). After filtering, all main results are confirmed

Tier	Lang	Full	Diff-proj	Δ	Rand Δ
HIGH	zh	.738	.746	+0.8	+0.4
HIGH	es	.745	.757	+1.2	+0.6
HIGH	de	.725	.743	+1.8*	+0.8
MID	hi	.670	.698	+2.8*	+0.8
MID	ru	.705	.727	+2.2*	+0.8
MID	tr	.660	.685	+2.5*	+0.8
MID	ko	.688	.682	-0.6	+0.3
LOW	sw	.598	.586	-1.2	+0.3
LOW	bn	.612	.647	+3.5*	+0.8

Table 11: Difficulty decomposition results (Qwen3-235B, 8-layer mean). Diff-proj = after projecting out difficulty direction. Rand Δ = random-direction control. * = $p < 0.05$.

Metric	Q-LR	Q-MLP	L-LR	L-MLP
EN AUROC	.810	.832	.770	.792
Mean ZS AUROC	.668	.638	.648	.622
Transfer gap	14.2	19.4	12.2	17.0
ZS pass rate	75%	55%	70%	50%

Table 12: LR vs MLP probe comparison (Q=Qwen3, L=Llama4). MLP overfits English-specific patterns, widening the transfer gap by $\sim$5 pp.

unchanged. This finding has field-level implications: **all cross-lingual probing studies should include selectivity checks**, especially for low-resource languages where probe capacity can mimic genuine signal.

4.11 Quantization Validation

We validate that INT4/W4A16 quantization does not introduce artifacts by comparing against FP16 at 5 layers per model on a 5% data subset. All $|\Delta\text{AUROC}| < 0.015$ and all CKA(FP16, quant) > 0.97 (Table 15).

The largest delta (-0.012 at Qwen3 L80) occurs at the deepest probed layer of the MoE model, where expert routing decisions may shift slightly under INT4. All deltas remain within bootstrap confidence intervals, supporting the validity of our quantized experimental setup.

4.12 Extended Language Coverage

We extend from 20 to 52 languages using GlobalMMLU and Belebele's full language sets (Table 16).

The rank correlation between 20-language and 52-language results is 0.95, confirming that our core analysis on 20 languages is representative. Very-low-resource languages ($<$ 1M training tokens) show 45% artifact rates and near-chance performance, reinforcing the necessity of probing controls. Writing-system clusters emerge as a moderate predictor of transfer quality within resource tiers.

5 Discussion

Practical recommendations. For practitioners deploying multilingual calibration: (1) select probe layer at 70–75% depth rather than the English-optimal layer; (2) use zero-shot transfer for HIGH-resource languages (ZS AUROC > 0.65); (3) apply linear projection with $\sim$80 parallel pairs for languages with ZS AUROC < 0.57; (4) for the ambiguous zone ($0.57-0.65$), validate on a small held-out set.

Metric	GlobalMMLU	Belebele	MMLU-ProX
EN AUROC	0.81	0.75	0.79
Mean ZS AUROC	0.668	0.632	0.658
ZS pass rate	75%	60%	72%
Pareto sig.	9	6	8
Lin. proj. Δ	+3.0	+2.5	+2.8
CACLR Δ	+0.3	+0.2	+0.3

Table 13: Cross-benchmark consistency (Qwen3-235B). All qualitative findings generalize. Cross-benchmark language ranking $\rho = 0.89$.

Depth	HIGH	MID	LOW	All
0–25%	15%	55%	80%	50%
25–50%	3%	28%	55%	29%
50–75%	0%	10%	18%	9%
75–100%	0%	2%	5%	2%
Overall	5%	22%	35%	22%

Table 14: Artifact rate (selectivity < 0.05) by resource tier and depth zone, averaged across 3 architectures. Low-resource shallow layers are predominantly artifacts.

Why distributional matching fails. The CACLR negative result is grounded in the theoretical analysis of Zhao et al. (2019): when $P(\text{correct} \mid \mathbf{h})$ differs across languages (the capability-specific component), marginal distribution matching cannot recover the conditional structure. Our CKA analysis provides empirical confirmation—MMD successfully reduces marginal distance while leaving the correctness subspace angle unchanged. Linear projection succeeds because it directly learns the subspace rotation without assuming distributional equivalence.

Limitations. Our study is limited to MCQA benchmarks, which enable clean correctness labeling but may not generalize to open-ended generation. The three architectures differ simultaneously in type (MoE vs dense), scale, layer count, and training data; architectural effects cannot be isolated ($n = 3$). Simpson's Paradox significance depends on the CKA depth profile shape, limiting its universality as a diagnostic. Probing controls flag 22% of language–layer combinations as artifacts, and we cannot rule out subtler capacity effects below our selectivity threshold.

6 Conclusion

We present the first large-scale study of cross-lingual correctness probe transfer across three architectures and 20 languages. The depth–transferability Pareto tradeoff is architecture-dependent, with CKA geometry providing explanatory power via a Simpson's Paradox that reverses the apparent negative CKA–transfer relationship. Distributional matching (CACLR, CORAL, high-dimensional MMD) uniformly fails—a systematic negative result explained by the inability of marginal matching to correct directional misalignment in the correctness subspace. Linear projection with $\sim$80 parallel pairs is the most effective label-free solution, particularly for low-resource languages, though it should be skipped when zero-shot transfer already works well. Our probing controls reveal that 22% of cross-lingual probing results are capacity artifacts, a finding with implications for the broader field. Together, these results validate that probe-based calibration extends to multilingual settings, while identifying when and why it succeeds or fails.

Model	Layer	FP16	Quant	Δ
Llama4	L8 (17%)	.635	.633	$-.002$
Llama4	L24 (50%)	.722	.717	$-.005$
Llama4	L44 (92%)	.770	.762	$-.008$
Qwen3	L16 (17%)	.632	.627	$-.005$
Qwen3	L52 (55%)	.742	.739	$-.003$
Qwen3	L80 (85%)	.810	.798	$-.012$

Table 15: Quantization validation (selected layers). Max $|\Delta| = 0.012$; CKA > 0.97 throughout. Quantization artifacts are negligible.

Tier	n_{20}	n_{52}	ZS AUROC	Art.%	Pass
HIGH	6	10	0.72	5%	100%
MID	8	15	0.66	20%	87%
LOW	6	16	0.58	32%	56%
V-LOW	0	11	0.53	45%	27%

Table 16: Language coverage expansion to 52 languages. Art.% = artifact rate. Pass = ZS AUROC > 0.58. Core findings replicate (rank correlation 20 vs. 52 = 0.95).

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